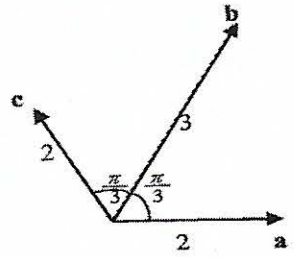


Vector 2 (Dot product)

- The vectors a , b and c have magnitude 2, 3 and 2 respectively and are directed as shown in the diagram.
 - Find $a \cdot b$, $b \cdot c$ and $c \cdot a$
 - If $x = a + 2b$ and $y = 2b - c$, find $x \cdot y$



- The vectors a and b are such that $|a| = 3$ and $|b| = 5$. Calculate the magnitude of $a + b$ and $a - b$ given that the angle between a and b is $\frac{\pi}{3}$.
Give the geometrical significance of the results.

- Given that $|\overrightarrow{AB}| = 3$, $|\overrightarrow{AC}| = 1$ and $\angle CAB = 60^\circ$ as shown in the figure.

Find

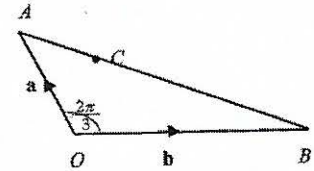
- $\overrightarrow{AB} \cdot \overrightarrow{AC}$
- $|\overrightarrow{AB} + 2\overrightarrow{AC}|$ and $|\overrightarrow{AB} - 2\overrightarrow{AC}|$



- In the figure, OAB is a triangle. C is a point on AB such that $AC:CB = 1:2$. Let $\overrightarrow{OA} = a$, $\overrightarrow{OB} = b$ and $\angle AOB = \frac{2\pi}{3}$.

If $|a| = 1$ and $|b| = 2$,

- Find $a \cdot b$.
- Express $\overrightarrow{OC}$ in term of a and b . Hence, find $|\overrightarrow{OC}|$.
- Let d be a vector such that $a + b + d = 0$, find $|d|$.



- Let a and b be two vectors such that $a = i + j$, $b = (c + 4)i + (c - 4)j$ and $\cos \theta = -\frac{3}{5}$, where θ is the angle between a and b . Find the value of the constant c .

- 3D ✱ 6. The angle between the vectors $i + j$ and $i + j + pk$ is $\frac{\pi}{4}$. Find the possible values of p .

- Let a and b be two vectors such that $a = 2i + 4j$, $|b| = \sqrt{5}$ and $\cos \theta = \frac{4}{5}$, where θ is the angle between a and b .

- Find $a \cdot b$.
- If $b = mi + nj$, find the values of m and n .

- Let a, b be two vectors such that $|a| = 4$, $|b| = 3$ and the angle between a and b is 60° .

- Find $a \cdot b$
- Find the value of k if the vectors $(a + kb) \cdot (a - 2b) = 0$.

- 3D ✱ 9. Given $a = i + 2j$ and $b = j + 2k$

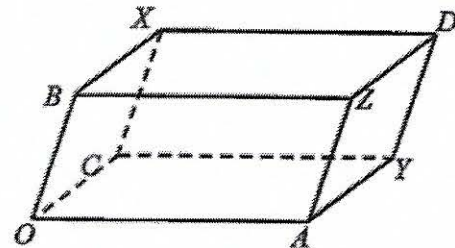
- Find
 - $|a|$ and $|b|$,
 - $a \cdot b$
- If $x = -2a + 3b$ and $y = 3a - 4b$, find $x \cdot y$.

- 3D ✱ 10. In the figure, the origin of position vectors is O , and points A, B, C

are given by $\overrightarrow{OA} = 5i$, $\overrightarrow{OB} = i + 3k$, $\overrightarrow{OC} = i + 4j$

A parallelepiped has OA, OB, OC as three edges, and the remaining vertices are X, Y, Z and D as shown.

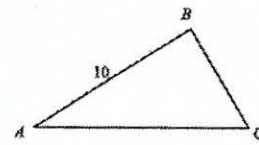
- Write down the vectors $\overrightarrow{OX}, \overrightarrow{OY}, \overrightarrow{OZ}$ and $\overrightarrow{OD}$ in terms of i, j, k .
- Calculate $\overrightarrow{OA} \cdot \overrightarrow{OD}$ and $\overrightarrow{OB} \cdot \overrightarrow{OY}$.



- Given $\overrightarrow{OA} = 4i + 3j$ and C is a point on OA such that $|\overrightarrow{OC}| = \frac{16}{5}$

- Find the unit vector in the direction of $\overrightarrow{OA}$. Hence find $\overrightarrow{OC}$.
- If $\overrightarrow{OB} = \lambda i + 4j$ and $\overrightarrow{BC} \cdot \overrightarrow{OA} = 0$, find the value of λ .

12. In the figure, $\angle B = 90^\circ$ and $|\overrightarrow{AB}| = 10$. Calculate $\overrightarrow{AB} \cdot \overrightarrow{AC}$.



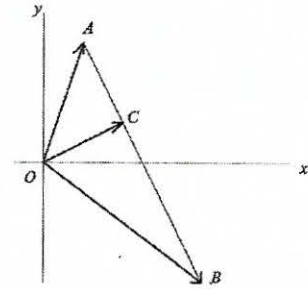
13. Given that $u = 3i + 2j$ and $v = 2i + \lambda j$, determine the values of λ so that

- a. u and v are at right angles,
b. u and v are parallel.

- c. The acute angle between u and v is $\frac{\pi}{4}$

14. Let $\overrightarrow{OA} = i + 3j$, $\overrightarrow{OB} = 4i - 3j$. C is a point on AB such that $\frac{AC}{CB} = r$.

- a. Express $\overrightarrow{OC}$ in terms of r, i and j .
b. Find the value of r if OC is perpendicular to AB .
Hence find $\overrightarrow{OC}$.



15. Let $\overrightarrow{OA} = i + 3j$, $\overrightarrow{OB} = 4i - j$, and C be a point dividing AB internally in the ratio $k:1$.

- a. Express $\overrightarrow{OC}$ in terms of k, i and j .
b. If OC is perpendicular to AB , find the value of k and $\overrightarrow{OC}$.

16. Given $\overrightarrow{OA} = 3i - 2j$, $\overrightarrow{OB} = i + j$. C is a point such that $\angle ABC$ is a right angle.

- a. Find $\overrightarrow{AB}$.
b. Find $\overrightarrow{AB} \cdot \overrightarrow{AB}$ and $\overrightarrow{AB} \cdot \overrightarrow{BC}$. Hence find $\overrightarrow{AB} \cdot \overrightarrow{AC}$.

17. The vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are given by $\overrightarrow{OP} = \lambda i + \mu j$, $\overrightarrow{OQ} = -\mu i + \lambda j$ where λ and μ are non-zero constants.

- a. Show that the angle between $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ is $\frac{\pi}{2}$ and find $\overrightarrow{PQ}$ in terms of λ and μ .
b. i. Given that R is the mid-point of PQ , express $\overrightarrow{OR}$ in terms of λ and μ .
ii. Show that OR is always perpendicular to PQ .

18. Let a, b be two vectors such that $a = 3i + 4j$ and $|b| = 4$. The angle between a and b is $\frac{\pi}{3}$.

- a. Find $a \cdot b$.
b. If the vector $(ma + b)$ is perpendicular to b , find the value of m and $(ma + b) \cdot a$.

19. P, Q and R are point on a plane such that $\overrightarrow{OP} = i + 2j$, $\overrightarrow{OQ} = 3i + j$ and $\overrightarrow{PR} = -3i - 2j$, where O is the origin.

- a. Find $\overrightarrow{PQ}$ and $|\overrightarrow{PQ}|$. Hence, find the value of $\cos \angle QPR$.
b. Find the value of $\sin \angle QPR$. Hence, find the area of ΔQPR .

Answers:

1. a. $a \cdot b = 3$, $b \cdot c = 3$, $c \cdot a = -2$; b. $x \cdot y = 38$

2. $|a + b| = 7$, $|a - b| = \sqrt{19}$. They represent the lengths of the diagonals of the parallelogram formed by a and b as two adjacent sides.

3. a. $\overrightarrow{AB} \cdot \overrightarrow{AC} = \frac{3}{2}$; b. $|\overrightarrow{AB} + 2\overrightarrow{AC}| = \sqrt{19}$, $|\overrightarrow{AB} - 2\overrightarrow{AC}| = \sqrt{7}$

4. a. $a \cdot b = -1$; b. $OC = \frac{2}{3}$; c. $|d| = \sqrt{3}$

5. $a \cdot b = 2c$, $c = -3$

6. $p = \frac{1}{2}\sqrt{2}$

7. a. $a \cdot b = 8$; b. $m = 2, n = 1; m = -\frac{2}{5}, n = \frac{11}{5}$

8. a. $a \cdot b = 6$; b. $k = \frac{1}{3}$

9. a. $|a| = \sqrt{5}$, $|b| = \sqrt{5}$, $a \cdot b = 2$; b. $x \cdot y = -56$

10. a. $\overrightarrow{OX} = 2i + 4j + 3k$, $\overrightarrow{OY} = 6i + 4j$, $\overrightarrow{OZ} = 6i + 3j$, $\overrightarrow{OD} = 7i + 4j + 3k$; b. $\overrightarrow{OA} \cdot \overrightarrow{OD} = 35$, $\overrightarrow{OB} \cdot \overrightarrow{OD} = 6$

11. a. $\overrightarrow{OC} = \frac{16}{5}(\frac{4}{5}i + \frac{3}{5}j)$; b. $\lambda = 1$

12. $\overrightarrow{AB} \cdot \overrightarrow{AC} = 100$

13. a. $\lambda = -3$, b. $\lambda = \frac{4}{3}$; c. $\lambda = -\frac{2}{5}$ or $\lambda = 10$

14. a. $OC = \frac{1+4r}{1+r}i + \frac{3-3r}{1+r}j$; b. $r = \frac{1}{2}$, $\overrightarrow{OC} = 2i + j$

15. a. $OC = \frac{1+4k}{1+k}i + \frac{3-k}{1+k}j$; b. $k = \frac{9}{16}$, $\overrightarrow{OC} = \frac{46}{25}i + \frac{39}{25}j$

16. a. $\overrightarrow{AB} = -2i + 3j$; b. $\overrightarrow{AB} \cdot \overrightarrow{AB} = 13$, $\overrightarrow{AB} \cdot \overrightarrow{BC} = 0$, $\overrightarrow{AB} \cdot \overrightarrow{AC} = 13$

17. a. $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 0$, $\overrightarrow{PQ} = -(\lambda + \mu)i + (\lambda - \mu)j$; b. $\overrightarrow{OR} = \frac{\lambda - \mu}{2}i + \frac{\mu + \lambda}{2}j$

18. a. $a \cdot b = 10$; b. $m = -\frac{8}{5}$, $(ma + b) \cdot a = -5$

19. a. $\overrightarrow{PQ} = 2i - j$, $|\overrightarrow{PQ}| = \sqrt{5}$, $\cos \angle QPR = -\frac{4\sqrt{65}}{65}$; b. $\sin \angle QPR = \frac{7}{\sqrt{65}}$, Area of $\Delta QPR = 7$